The Exponential Generating Function

For a sequence $a_0, a_1, a_2, \cdots$ of real numbers,

$$f(x) = a_0 + a_1 x + a_2 \frac{x^2}{2!} + a_3 \frac{x^3}{3!} + \cdots$$

$$= \sum_{r=0}^{\infty} a_r \frac{x^r}{r!}$$

is called the exponential generating function for the given sequence.

Note :- If $f(x)$ is the E.g.f for a sequence $a_0, a_1, a_2, \cdots$ then $f(x)$ is an ordinary generating function for the sequence $a_0, a_1, \frac{a_2}{2!}, \frac{a_3}{3!}, \cdots$.

i.e $f(x) = a_0 + a_1 x + a_2 \frac{x^2}{2!} + a_3 \frac{x^3}{3!} + \cdots$ → Exponentia
for the seque
$a_0, a_1, a_2, \cdots$

$f(x) = a_0 + a_1 x + \frac{a_2}{2!} x^2 + \frac{a_3}{3!} x^3 + \cdots$

→ ordinary g.f for the sequence
$a_0, a_1, \frac{a_2}{2!}, \frac{a_3}{3!}, \cdots$ (E.g.f)

(1). Find the Exponential generating function for the sequence $1, 1, 1, 1, \cdots$

▲ We have $f(x) = a_0 + a_1 x + a_2 \frac{x^2}{2!} + a_3 \frac{x^3}{3!} + \cdots$ as an

Given that $a_0 = 1, a_1 = 1, a_2 = 1, a_3 = 1, \cdots$

$f(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = e^x$

(2). **Note :-** e^x is the o.g.f for the sequence $1, 1, \frac{1}{2!}, \cdots$
Find the E.g.f for the sequence $0, 1, 1, 1, \cdots$

▲ $f(x) = x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

(3) Find the E.g.f for the sequence $1, -1, 1, -1, 1, -1, \cdots$

▲ $f(x) = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \frac{x^5}{5!} + \cdots = e^{-x}$

4.

⑥. Determine the sequence generated by each of the following exponential generating functions:

(i) $f(x) = 3e^{3x}$ (ii) $f(x) = 5e^{5x}$ (iii). $f(x) = 6e^{5x} - 3e^{2x}$

(iv) $f(x) = 7e^{8x} - 4e^{3x}$ (v) $f(x) = e^x + x^2$ (vi) $f(x) = 2e^x + 3x^2$

(vii) $f(x) = e^{2x} - 3x^3 + 5x^2 + 7x$ (viii). $f(x) = e^{3x} - 28x^3 - 6x^2 + 9x$

(ix). $f(x) = \frac{1}{1-x}$ (x) $f(x) = \frac{3}{1-2x} + e^x$.

(i) $f(x) = 3e^{3x} = 3\sum_{r=0}^{\infty} \frac{[3x]^r}{r!} = 3\left[1 + 3\frac{x}{1!} + 3^2\frac{x^2}{2!} + \cdots\right] = \left[3 + 3^2x + 3^3\frac{x^2}{2!} + \cdots\right]$

Note: $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{r=0}^{\infty} \frac{x^r}{r!}$

$f(x)$ is the Exponential generating function for the sequence

$3, 3^2, 3^3, \cdots$

(ii) $f(x) = 5e^{5x} = 5\sum_{r=0}^{\infty} \frac{[5x]^r}{r!} = 5\left[1 + 5\frac{x}{1!} + 5^2\frac{x^2}{2!} + \cdots\right] = 5 + 5^2\frac{x}{1!} + 5^3\frac{x^2}{2!} + \cdots$

$f(x)$ is the E.g.f for the sequence $5, 5^2, 5^3, \cdots$

(iii). $f(x) = 6e^{5x} - 3e^{2x}$

$= 6\sum_{r=0}^{\infty} \frac{[5x]^r}{r!} - 3\sum_{r=0}^{\infty} \frac{[2x]^r}{r!}$

$= 6\left[1 + 5\frac{x}{1!} + 5^2\frac{x^2}{2!} + \cdots + 5^n\frac{x^n}{n!} + \cdots\right]$

$- 3\left[1 + 2\frac{x}{1!} + 2^2\frac{x^2}{2!} + \cdots + 2^n\frac{x^n}{n!} + \cdots\right]$

$= 3 + 24\frac{x}{1!} + 138\frac{x^2}{2!} + \cdots \left[6(5^n) - 3(2^n)\right]\frac{x^n}{n!} + \cdots$

$f(x)$ is the E.g.f for the sequence,

$3, 24, 138, \cdots \left[6[5^n] - 3[2^n]\right] \cdots$

4, +, 2

7). Using the exponential generating function, find the no. of ways in which 4 of the letters in the word ENGINE be arranged.

we have E - 2, N - 2, G - 1, I - 1.

In an arrangement of these letters, E can appear 0, 1 or 2 times. The Exponential generating function corresponding to the letter E is

$$f_1(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!}$$

Illy, the E.g.f corresponding to the letter N is

$$f_2(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!}$$

Since G can appear 0 or 1 time in an arrangement. Then the corresponding Exponential generating function is

$$f_3(x) = 1 + \frac{x}{1!}$$

Also, I can appear 0 or 1 time in an arrangement.

$$f_4(x) = 1 + \frac{x}{1!}$$

$\frac{4!}{2!} \cdot 4! \left[2 + 2 + \frac{4}{4!}\right]$

(8) Using the exponential generating function, find the no. of ways in which 4 of the letters in the word "HAWAII".

A we have: A → 2, I → 2, H - 1, W - 1.

In an arrangement of these letters, A can appear 0, 1 times. Then the corresponding Exponential generating function is

$$f_1(x) = 1 + x + \frac{x^2}{2!}$$

Illy the letter I can appear 0, 1 or 2 times. Then the corresponding Exponential generating function is

$$f_2(x) = 1 + x + \frac{x^2}{2!}$$

Since H and W are appear 0 or 1 times. Then the corresponding ~~corresp~~ Exponential generating functions are:

$$f_3(x) = 1+x \quad \text{and} \quad f_4(x) = 1+x.$$

Then $f(x) = f_1(x)\, f_2(x)\, f_3(x)\, f_4(x)$

$$= \left[1+x+\frac{x^2}{2!}\right] \times \left[1+x+\frac{x^2}{2!}\right] \times [1+x][1+x]$$

$$= \left[1+2x+2x^2+x^3+\frac{x^4}{4}\right][1+2x+x^2]$$

$$= \left[1+2x+2x^2+x^3+\frac{x^4}{4}\right][1+2x+x^2+0x^3+0x^4]$$

We have: $x^4\left[1\times 0 + 2\times 0 + \text{[illegible]} + 2 + 2 + \frac{1}{4}\right]$

$$= x^4\left[2+2+\frac{1}{4}\right] = \frac{x^4}{4!} \times 4!\left[2+2+\frac{1}{4}\right]$$

The coefficient of $\frac{x^4}{4!}$ is $4!\left[2+2+\frac{1}{4}\right] = 102$.

(9). Find the Exponential generating function for the number of ways to arrange 'n' letters, $n \geq 0$ selected from each of the following words:

(i) HAWAII (ii) MISSISSIPPI (iii) ISOMORPHISM

▲ (i) We have: A → 2, I → 2, H → 1 : W → 1.

The corresponding Exponential generating function is

$$f(x) = \left[1+x+\frac{x^2}{2!}\right]\left[1+x+\frac{x^2}{2!}\right][1+x][1+x]$$

$$= \left[1+x+\frac{x^2}{2!}\right]^2 [1+x]^2$$

(ii) We have: $I \to 4,\ S \to 4,\ P \to 2,\ M \to 1$

Therefore the required Exponential generating function

is $f(x) = \left[1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}\right] \times \left[1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}\right] \times \left[1 + x + \frac{x^2}{2!}\right] \times [1 + x]$

$$= \left[1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}\right]^2 \left[1 + x + \frac{x^2}{2!}\right][1 + x]$$

(iii) We have: $I \to 2,\ S \to 2,\ O \to 2,\ M \to 2$

$R \to 1,\ P \to 1,\ H \to 1$

Therefore, the required Exponential generating function

is $f(x) = \left[1 + x + \frac{x^2}{2!}\right] \times \left[1 + x + \frac{x^2}{2!}\right] \times \left[1 + x + \frac{x^2}{2!}\right] \times \left[1 + x + \frac{x^2}{2!}\right]$

$$\times [1 + x] \times [1 + x] \times [1 + x]$$

$$f(x) = \left[1 + x + \frac{x^2}{2!}\right]^4 [1 + x]^3 =$$

(10). A ship carries 48 flags, 12 each of the colors red, white, blue and black. If 12 of these flags are placed on a vertical pole in order to communicate a signal to other ships.

(a) How many of these signals use an Even number of blue flags and an odd number of black flags?

(b) How many of ~~these~~ the signals have at least three white flags or no white flag at all?

In this case, the problem is to find the no. of arrangements of 12 flags of which blue flag can appear Even no. of times and black flag can appear odd no. of times and there is no restriction on others.

Corresponding to blue flags, which are to be Even in number, we have

$$f_1(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots = \frac{1}{2}\left[e^x + e^{-x}\right]$$

Corresponding to black flags, which are to be odd in number, we have:

$$f_2(x) = \frac{x}{1!} + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots = \frac{1}{2}\left[e^x - e^{-x}\right]$$

Since there are no restrictions on red and white flags, corresponding to these, we have:

$$f_3(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = e^x \text{ and}$$

$$f_4(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = e^x.$$

Then $f(x) = f_1(x)\, f_2(x)\, f_3(x)\, f_4(x)$

$$= \frac{1}{2}\left[e^x + e^{-x}\right] \times \frac{1}{2}\left[e^x - e^{-x}\right] \times e^x \times e^x$$

$$= \frac{1}{4}\left[e^{2x} - e^{-2x}\right] e^{2x}$$

$$= \frac{1}{4}\left[e^{4x} - 1\right] = \frac{1}{4}\left[\sum_{r=0}^{\infty} \frac{4^r x^r}{r!} - 1\right]$$

The coefficient of $\frac{x^{12}}{12!}$ is $\frac{1}{4}\left[4^{12}\right] = 4^{11} = 4194304$.

This is the required number.

(b) In this case, we have

$$f(x) = \left[x^0 + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots\right]\left[1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right]^3$$

$$= \left[e^x - x - \frac{x^2}{2!}\right]\left[e^x\right]^3$$

$$= \left[e^x - x - \frac{x^2}{2!}\right] e^{3x} = e^{4x} - xe^{3x} - \frac{x^2}{2} e^{3x}$$

$$f(x) = \sum_{r=0}^{\infty} \frac{(4x)^r}{r!} - x \sum_{r=0}^{\infty} \frac{(3x)^r}{r!} - \frac{x^2}{2} \sum_{r=0}^{\infty} \frac{(3x)^r}{r!}$$

$$= \sum_{r=0}^{\infty} \frac{4^r x^r}{r!} - \sum_{r=0}^{\infty} 3^r \frac{x^{r+1}}{r!} - \frac{1}{2} \sum_{r=0}^{\infty} 3^r \frac{x^{r+2}}{r!}$$

The coefficient of $\frac{x^{12}}{12!}$ is

$$= 4^{12} - [3^{11} \times 12] - \frac{1}{2}[3^{10} \times 12 \times 11]$$

$$= 10754218.$$

This is the required number.

A company hires 11 new employees, each of whom is to be assigned to one of four subdivisions. Each subdivision will get at least one new employee. In how many ways can these assignments be made?

Let A, B, C, D denote the four subdivisions of the company. Then the problem is equivalent to counting the no. of sequences of length 11 in which at least one of A, B, C, D occurs. Since in every sequence of the desired type each of A, B, C, D occurs at least once, we take the Exponential generating function as

$$f(x) = \left[x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots\right]^4$$

$$= [e^x - 1]^4$$

$$= e^{4x} - 4e^{3x} + 6e^{2x} - 4e^x + 1$$

$$= \sum_{r=0}^{\infty} \frac{4^r x^r}{r!} - 4 \sum_{r=0}^{\infty} \frac{3^r x^r}{r!} + 6 \sum_{r=0}^{\infty} \frac{2^r x^r}{r!} - 4 \sum_{r=0}^{\infty} \frac{x^r}{r!} + 1$$

Partitions of Integers :

Partitions of a the integer 'n' is a collection of the integer parts with sum equal to 'n'. The no. of partitions of an integer 'n' is denoted by $p(n)$.

Ex:- $1 \longrightarrow 1$ $\quad \therefore p(1) = 1$

$2 \longrightarrow 2, 1+1$ $\quad \therefore p(2) = 2$

$3 \longrightarrow 3, 2+1, 1+1+1$ $\quad \therefore p(3) = 3$

$4 \longrightarrow 4, 3+1, 2+2, 2+1+1, 1+1+1+1$ $\quad \therefore p(4) = 5$

$5 \longrightarrow 5, 4+1, 3+2, 3+1+1, 2+2+1, 2+1+1+1, 1+1+1+1+1$ $\quad \therefore P(5) = 7$.

We should like to obtain $p(n)$ for a given 'n' without having to list all the partitions. We need a tool to keep track of the number of 1's, 2's, 3's --- , n's that are used as summands for 'n'.

If $n \in Z^+$, the no. of 1's we can use is 0, or 1 or 2 or ---

The Power series repreention of the abale case is:

$x^1 \longrightarrow [x^1]^0 + [x^1]^1 + [x^1]^2 + [x^1]^3 + \cdots = 1 + x + x^2 + x^3 + \cdots = \frac{1}{1-x}$

The no. of 2's we can use is 0 or 1 or 2 or -----

The Power series representation of the abale case can be written as:

$x^2 \rightarrow [x^2]^0 + [x^2]^1 + [x^2]^2 + \cdots = 1 + x^2 + x^4 + \cdots = \frac{1}{1-x^2}$

The no. of 3's we can use is 0 or 1 or 2 or --- and hence the Power series can be written as:

$x^3 \rightarrow [x^3]^0 + [x^3]^1 + [x^3]^2 + \cdots = 1 + x^3 + x^6 + \cdots = \frac{1}{1-x^3}$

. . . . . .

The no. of n's, we can use is 0 or 1 or 2 or ---

$x^n \rightarrow [x^n]^0 + [x^n]^1 + [x^n]^2 + \cdots = 1 + x^n + x^{2n} + \cdots = \frac{1}{1-x^{2n}}$

The generating function function for the no. of partition of any the integer 'n' is

$$P(n) = f(x) = \frac{1}{1-x}\,\frac{1}{1-x^2}\,\frac{1}{1-x^3} \times \cdots \times \frac{1}{1-x^n} = \prod_{k=1}^{\infty} \frac{1}{1-x^k} = \sum_{n=0}^{\infty} \text{Or}$$

$$= \prod_{k=1}^{\infty} [1-x^k]^{-1} = \prod_{k=1}^{\infty} \sum_{\gamma=0}^{\infty} x^{\gamma k}. \quad \text{——} \quad \text{——} (1)$$

Note: (1). There is no simple formula for $P(n)$, further the coefficient of x^n is $P(n)$.

In order to determine $p(3)$, we have to find the coefficient of x^3 in ①.

$$f(x) = \prod_{k=1}^{\infty} \frac{1}{1-x^k} = \frac{1}{1-x} \times \frac{1}{1-x^2} \times \frac{1}{1-x^3} \times \cdots$$

$$= [1+x+x^2+x^3+\cdots][1+x^2+x^4+\cdots][1+x^3+\cdots]\cdots$$

$$= [1+x+x^2+x^3+\cdots][1+x^2+\cdots][1+x^3+\cdots]\cdots$$

$$= [1+x+x^2+x^3+\cdots][1+x^2+x^3+x^5+\cdots]\cdots$$

$$= [1+x+x^2+x^3+\cdots][1+x^2+x^3+\cdots]\cdots$$

$$= [1+x+x^2+x^3+\cdots][1+0x+x^2+x^3+\cdots]\cdots$$

$$= 1+x+[1+0+1]x^2+[1+1+0+1]x^3+\cdots$$

$$= 1+x+2x^2+3x^3+\cdots$$

$p(3) = 3$.

② For example, $p(4)$ is the coefficient of x^4 in

$$f(x) = \prod_{k=1}^{\infty} \frac{1}{1-x^k} = \frac{1}{1-x} \times \frac{1}{1-x^2} \times \frac{1}{1-x^3} \times \frac{1}{1-x^4} \times \cdots$$

$$= [1 + x + x^2 + x^3 + x^4 + \cdots][1 + x^2 + x^4 + \cdots][1 + x^3 + x^6 + \cdots][1 + x^4 + \cdots]$$

$$= [\text{illegible}]$$

$$= 1 + x + 2x^2 + 3x^3 + 5x^4 + \cdots$$

The coefficient of x^4 is 5.

$\therefore \quad p(4) = 5.$

For $n = 6$, we have the following partitions:

6

5+1

4+2, 4+1+1

3+3, 3+2+1, 3+1+1+1

2+2+2, 2+2+1+1, 2+1+1+1+1

1+1+1+1+1+1

$\therefore p(6) = 11,$

out of 11 partitions of 6, the following partitions represents the no. of partitions of 6 into distinct parts (or) distinct summands:

6, 5+1, 4+2, 3+2+1

The no. of partitions of a the integer 'n' into distinct parts (or) distinct summands can be denoted by $p_d(n)$.

$$\therefore p_d(6) = 4$$

out of 11 partitions of 6, the partition represents the no. of partitions of 6 into odd summands and it is denoted $p_o(6)$.

5+1, 3+3, 3+1+1+1, 1+1+1+1+1+1

$$\therefore \quad p_o(6) = 4$$

Find the generating function for $P_d(n)$, the no. of partitions of a the integer 'n' into distinct summands.

▲ For any the integer k, the power series is

$$[x^k]^0 + [x^k]^1$$

$$\therefore f(x) = [x^1]^0 + [x^1]^1 \left[[x^2]^0 + [x^2]^1\right]\left[[x^3]^0 + [x^3]^1\right]\cdots$$

$$= [1+x][1+x^2][1+x^3]\cdots$$

$$\therefore f(x) = \prod_{k=1}^{\infty} [1+x^k] = P_d(x)$$

$P_d(n)$ = the coefficient of x^n in $[1+x][1+x^2]\cdots[1+x^n]$.

Note :- $P_d(0) = 1$.

② Find the generating function for $P_o(n)$, the no. of partitions of n into odd summands. Hence show that $P_o(n) = P_d(n)$.

▲ $1 \to [x^1]^0 + [x^1]^1 + [x^1]^2 + \cdots = 1 + x + x^2 + \cdots = \frac{1}{1-x}$

$3 \to [x^3]^0 + [x^3]^1 + [x^3]^2 + \cdots = 1 + x^3 + x^6 + \cdots = \frac{1}{1-x^3}$

$5 \to [x^5]^0 + [x^5]^1 + [x^5]^2 + \cdots = 1 + x^5 + x^{10} + \cdots = \frac{1}{1-x^5}$.

$$P_o(x) = \frac{1}{1-x} \times \frac{1}{1-x^3} \times \frac{1}{1-x^5} \times \cdots = \prod_{k=1}^{\infty} \frac{1}{1-x^{2k-1}} \quad -①$$

$P_o(n)$ is the coefficient of x^n in ①.

Now we have to show that $P_o(n) = P_d(n)$.

$$1+x = \frac{1-x^2}{1-x}, \quad 1+x^2 = \frac{1-x^4}{1-x^2}, \quad 1+x^3 = \frac{1-x^6}{1-x^3}, \cdots$$

we have $P_d(x) = [1+x][1+x^2][1+x^3][1+x^4]\cdots$

$$= \frac{1-x^2}{1-x} \times \frac{1-x^4}{1-x^2} \times \frac{1-x^6}{1-x^3} \times \frac{1-x^8}{1-x^4} \times \cdots$$

$$= \frac{1}{1-x} \times \frac{1}{1-x^3} \times \frac{1}{1-x^5} \times \cdots = P_o(x)$$

From the Equality of the generating function $P_d(n) = P_o(n)$, $\forall n \geq 0$.

③ Find the generating function for the no. of partitions of each Summands must appear an Even no. of times and find the generating function where each Summand must odd and must appear an odd number of times.

$1 \to [x^1]^0 + [x^1]^2 + [x^1]^4 + \cdots = 1 + x^2 + x^4 + \cdots = \frac{1}{1-x^2}$

$2 \to [x^2]^0 + [x^2]^2 + [x^2]^4 + \cdots = 1 + x^4 + x^8 + \cdots = \frac{1}{1-x^4}$

$3 \to [x^3]^0 + [x^3]^2 + [x^3]^4 + \cdots = 1 + x^6 + x^{12} + \cdots = \frac{1}{1-x^6}$

$\therefore P_e[x] = \prod_{k=1}^{\infty} \frac{1}{1-x^{2k}}$

The coefficient of x^n represents $P_e[n]$ [i.e the no. of partitions of n where each Summands must appear an Even no. of time.

$1 \to [x^1]^1 + [x^1]^3 + [x^1]^5 + \cdots = x + x^3 + x^5 + \cdots = x[1 + x^2 + x^4 + \cdots] = \frac{x}{1-x^2}$

$3 \to [x^3]^1 + [x^3]^3 + [x^3]^5 + \cdots = x^3[1 + x^6 + x^{12} + \cdots] = \frac{x^3}{1-x^6}$

$5 \to [x^5]^1 + [x^5]^3 + [x^5]^5 + \cdots = x^5[1 + x^{10} + x^{20} + \cdots] = \frac{x^5}{1-x^{10}}$

Generating function of the required type

$$= \frac{x}{1-x^2} \times \frac{x^3}{1-x^6} \times \frac{x^5}{1-x^{10}} \times \cdots = \prod_{k=1}^{\infty} \frac{x^{2k-1}}{1-x^{2[2k-1]}}$$

④ Show that the no. of partitions of a +ve integer 'n' where no Summand appears more than twice is Equal to the no. of partition of 'n', where no Summand is divisible by '3'.

$1 \to [x^1]^0 + [x^1]^1 + [x^1]^2 = 1 + x + x^2 = \frac{1-x^3}{1-x}$

$2 \to [x^2]^0 + [x^2]^1 + [x^2]^2 = 1 + x^2 + x^4 = \frac{1-x^6}{1-x^2}$

$3 \to [x^3]^0 + [x^3]^1 + [x^3]^2 = 1 + x^3 + x^6 = \frac{1-x^9}{1-x^3}$

$P_{NTM2}(n) = \frac{1-x^3}{1-x} \times \frac{1-x^6}{1-x^2} \times \frac{1-x^9}{1-x^3} \times \cdots = \prod_{k=1}^{\infty} \frac{1-x^{3k}}{1-x^k}$

$= \frac{1}{1-x} \times \frac{1}{1-x^2} \times \frac{1}{1-x^4} \times \frac{1}{1-x^5} \times \frac{1}{1-x^7} \times \cdots$

no Summand is